

src/abivaldate

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This module provides a command-line tool for validating ABI sequences against a reference FASTA file using Smith-Waterman alignment and variant calling.

The abivaldate tool loads reference sequences from a FASTA file, then aligns each ABI sequence against the references to find the best match. If no suitable alignment is found in the forward orientation, it automatically tries the reverse complement. It performs variant calling and outputs results in VCF format.

Command-line usage:

```
abivaldate -r reference.fasta -o output.vcf file1.ab1 file2.ab1 ...
```

Options:

-r, --reference FILE	Reference FASTA file (required)
-o, --output FILE	Output VCF file (required)
-h, --help	Show help message
--version	Show version information
--min-identity FLOAT	Minimum alignment identity (default: 0.80)
--min-coverage FLOAT	Minimum coverage of ABI sequence (default: 0.90)
--score-match INT	Score for match (default: 10)
--score-mismatch INT	Score for mismatch (default: -8)
--score-gap INT	Score for gap (default: -10)
-v, --verbose	Verbose output
-t, --table	Output summary table

1 Imports

abif